ai:D

Christian A. Klöckner// 2026-10-14

Uncertainty and trust in AI systems from a psychological perspective

agenda



What is trust?

Trust between human actors
Trust in the AI context

Measures to increase trust in AI

Aspects that increase the trust in AI systems and their communication

What is uncertainty?

Psychological definitions and effects of uncertainty

Communicating uncertainty

How to communicate about uncertain results / predictions

What does this mean for aiD?

SECTION 1

What is trust?

SECTION 1 // What is trust?

Trust between human actors

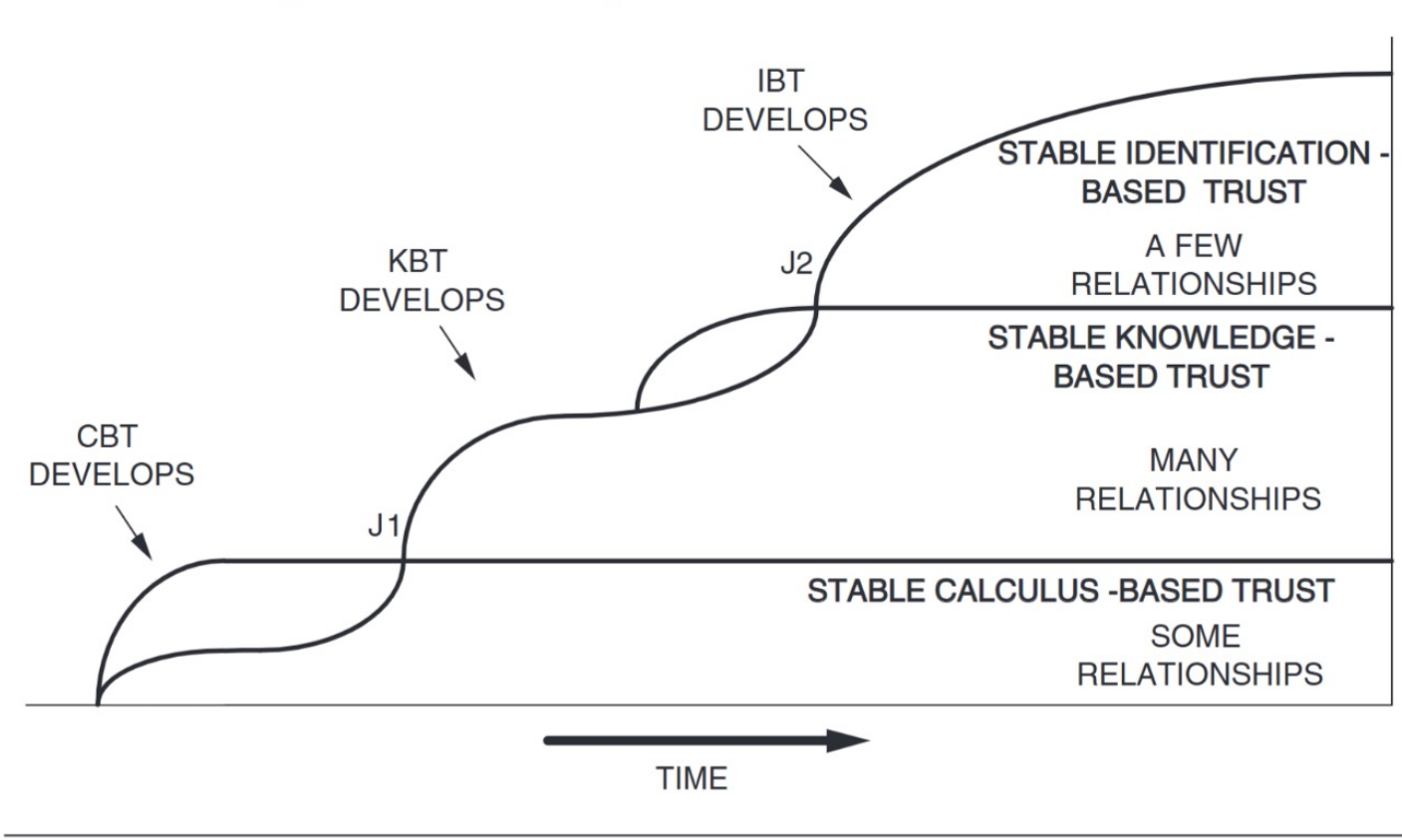
Trust in the AI context

Trust between humans

- Trust between humans needs to be built over time and is created in interactions
- Trust is often treated to have three dimensions
  - Trust in competence
  - Trust in integrity
  - Trust in benevolence
- Elements that create trust between humans
  - Consistency and reliability
  - Transparency and honesty
  - Integrity and ethic principles
  - Vulnerability and reciprocity
  - Empathy and care



The Stages of Trust Development—Lewicki & Bunker (1995, 1996)





Source: Adapted from Lewicki and Bunker (1995).
Note: J1 = At this juncture, some calculus-based trust relationships become knowledge-based trust relationship
J2 = at this juncture, a few knowledge-based trust relationships where positive affect is present go on to becom
identification-based trust relationships.

The trust equation

$$T = \frac{C\text{redibility} + R\text{eliability} + I\text{ntimacy}}{S\text{elf-Orientation}}$$

Trustworthiness



T stands for Trustworthiness—how much the buyer/client trusts the seller, or consultant.

C is Credibility—it speaks to words. Can we be believed, based on our credentials and truthfulness?

R is Reliability—it speaks to the consistency of actions. Do we meet expectations, and do our actions reflect our words (integrity)?

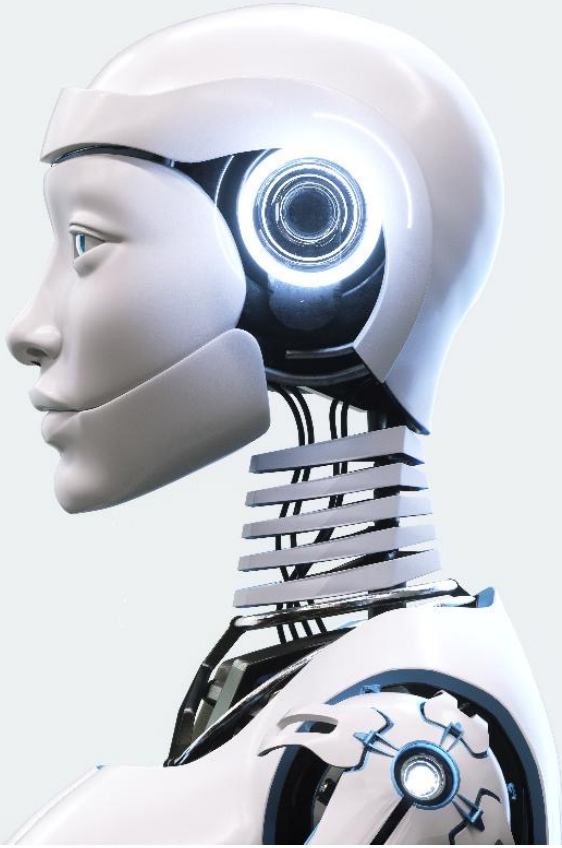
I is Intimacy—it speaks to the security or safety a client feels. Do we connect with and understand them? Do they feel safe sharing information with us?

S is Self-Orientation, the lone term in the denominator, and it has a double meaning. Partly it's about selfishness. Are we client-centric for the sake of the client? Or client-centric like a vulture?

https://trustedadvisor.com/articles/the-trust-equation-a-primer

Trust in the AI context

- Here trust is differentiated into
  - Cognitive trust -> trusting the results of an AI system
  - Emotional trust -> emotional relation to the AI system
- Cognitive trust is impacted by
  - Form of AI (robot, virtual, embedded)
  - its intelligence (as in capabilities)
  - tangibility, transparency, reliability, immediacy
  - transparency, explicability, accuracy, reliability,
- Emotional trust
  - AI’s degree of anthropomorphism, embodiment
  - automation (the degree to which AI can take over decisions)
- Other elements of trust in AI
  - mass data extraction



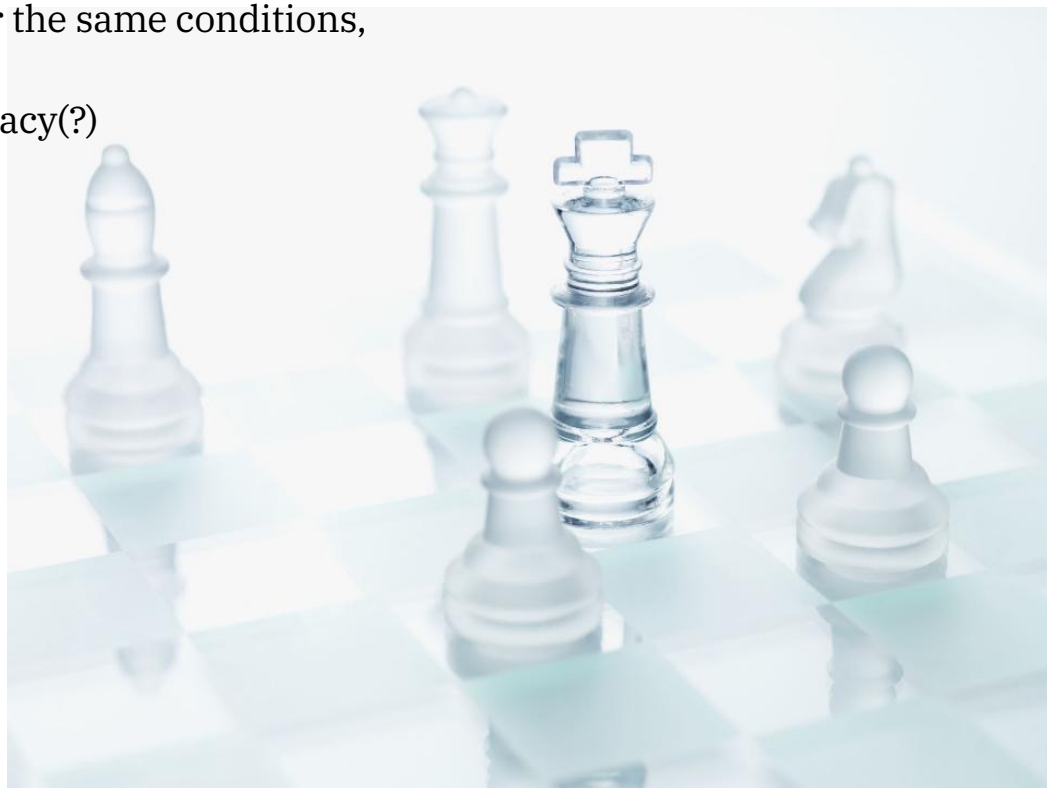
SECTION 2

Measures to increase trust in AI

Aspects that increase the trust in AI systems and their communication

Increasing trust in AI systems

- Produce credible results with transparency
- Be consistent and reliable (produce the same results under the same conditions, be predictable)
- Protect the user (data and person), create a feeling of intimacy(?)
- Build a personal relation(?)
- Do we need antromorph systems? For whom? When?



SECTION 3

What is uncertainty?

Psychological definitions and effects of uncertainty

Psychological definitions

Article	Definition of uncertainty
Dağtas and Sehnaz Sahinkarakas (2024)	"The subjective feeling of doubt and not knowing how to decode the present, the future, and the past (Jordan and McDaniel, 2014)". "In the present study, which is part of a comprehensive Ph.D. research, uncertainty is described as a psychological, personal, subjective experience of not being certain and not being able to make an accurate prediction concerning individual aspects, other people, and the features of the context."
Alquist et al. (2020)	"Uncertainty involves an individual lacking important information (Bar-Anan et al., 2009)."
Bar-Anan et al. (2009)	"Uncertainty refers to the state of an organism that lacks information about whether, where, when, how, or why an event has occurred or will occur (Knight, 1921)."
Salah et al. (2023)	Page 1: "Uncertainty reflects a lack of confidence in one's ability to predict particular outcomes (Penrod, 2001), and it may exact a significant burden on emotional and mental wellbeing (Eastwood et al., 2008; Eche et al., 2019; Perez et al., 2020)."
Alquist and Baumeister (2024)	Page 1: "Uncertainty is a cognitive-emotional state of not knowing which of multiple possibilities is or will be true (FeldmanHall and Shenhav, 2019)."
van Horen and Millet (2022)	Page 2: "Uncertainty due to external, unpredictable events (e.g., pandemics,



Franssen, M., Verstegen, R., Bazilinskyy, P., van der Spek, E. D., & Martens, M. H. (2026). Psychological uncertainty in human experience and behavior: a systematic review on measurements, induction, and effects. *Frontiers in Psychology*, *17*, 1844980.

Statistical definitions / AI definitions

For a machine learning model, predictive uncertainty is simply defined as the uncertainty of its predictions. It is caused by the model, i.e. epistemic uncertainty, or the data, i.e. aleatoric uncertainty (Kendall and Gal 2017, Hüllermeier and Waegeman 2021). The first one can be due to distribution shifts occurring on real-world data, or to errors in the model design (e.g. too large neural network overfitting the data) or in the training procedure. This type of uncertainty can be reduced by modifying the model architecture, learning process, or by acquiring more training data representative of real-world distributions. On the other hand, the aleatoric uncertainty cannot be reduced and can be caused by errors or noise in the measurement of inputs or labels, such as errors in labeling causing class overlap (Kendall and Gal 2017, Gawlikowski et al. 2023).

Bonnier, T. (2025). *Estimating and Explaining Predictive Uncertainty in Machine Learning* (Doctoral dissertation, Sorbonne Université).

Statistical definitions / AI definitions

It is worth noting that there are different degrees of uncertainty from certainty to ignorance. As a situation of inadequate information, uncertainty could correspond to inexactness, unreliability, and ignorance (Funtowicz and Ravetz 1990). A distinction can be made between uncertainties that can be managed through probabilities and uncertainties that cannot (Morgan et al. 1990). Improbable and unpredicted catastrophic events belong to the category of recognized ignorance (Walker et al. 2013) and are called “black swans” by Taleb (2007): e.g. 2007 mortgage crisis; 2011 earthquake in Japan.

Bonnier, T. (2025). *Estimating and Explaining Predictive Uncertainty in Machine Learning* (Doctoral dissertation, Sorbonne Université).

Psychological effects of uncertainty

- High uncertainty might create low trust
  - Low reliability
  - Low competency
  - Low predictability
- Not being open about uncertainty is at odds with transparency rule
- Uncertainty triggers decision biases
  - Anchoring heuristics (e.g., Furnham & Boo, 2011)
  - Optimistic bias (e.g., Sharot, 2011)
  - Framing effects (e.g., Tversky and Kahneman, 1981)



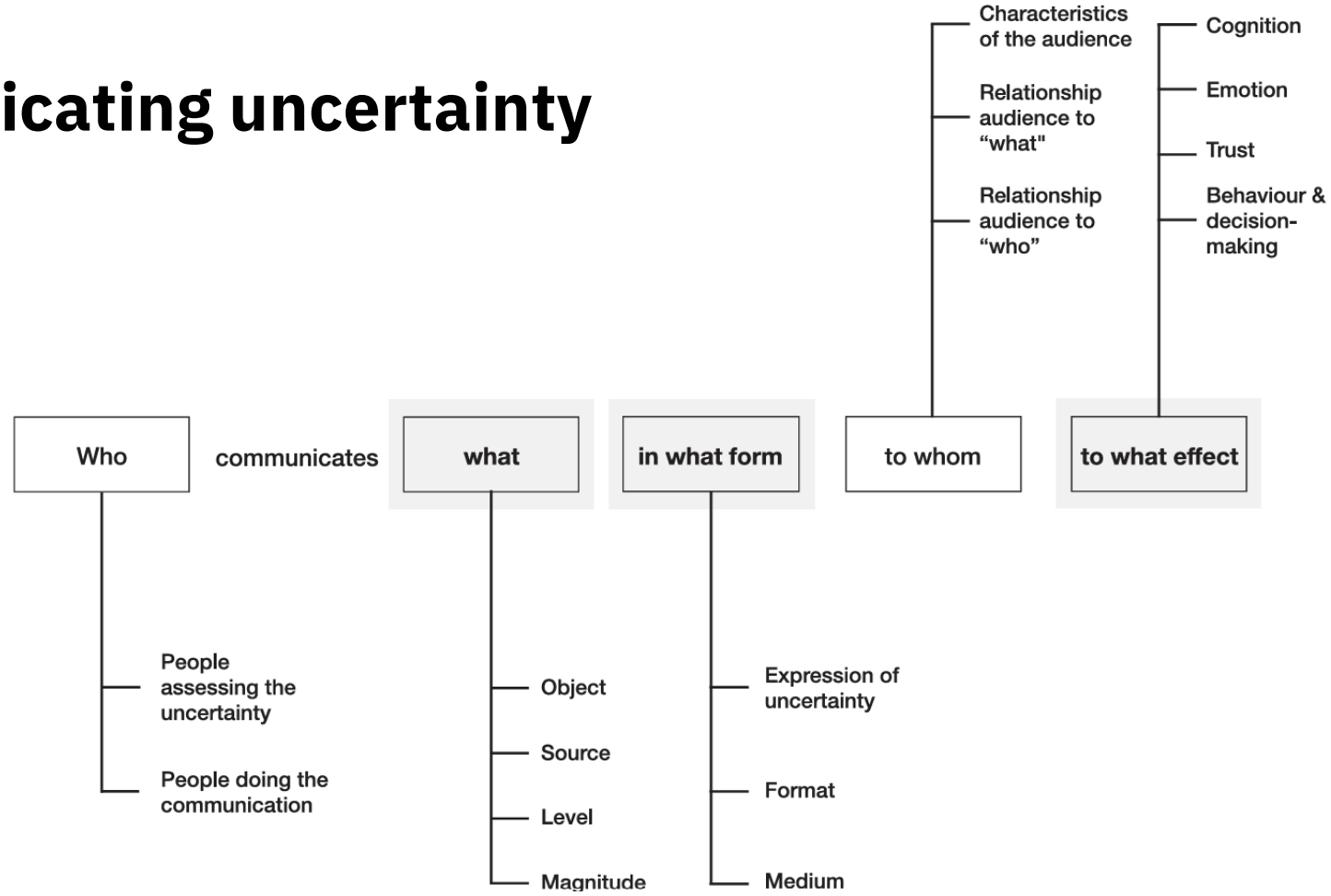
SECTION 4

Communicating uncertainty

SECTION 4 // What is trust?

How to communicate about uncertain results / predictions

Communicating uncertainty



Van Der Bles, A. M., Van Der Linden, S., Freeman, A. L., Mitchell, J., Galvao, A. B., Zaval, L., & Spiegelhalter, D. J. (2019). Communicating uncertainty about facts, numbers and science. *Royal Society open science*, *6*(5), 181870.

Communicating uncertainty

Box 5(*b*). Advice to researchers of how to communicate uncertainty around evidence.

When designing your studies, it would be useful to clearly identify and state:

— Is the uncertainty you are studying about the past/present or about the future—and is it likely to be perceived as past/present or future by the audience?
— What are the psychological effects you are studying? *Cognition* (including understanding), *emotional* (affect), *trust* or *behaviour* (including decision-making)?
— What are you varying about the uncertainty itself? (The *object*—is it a categorical or continuous object or is it a hypothesis; the *source* of the uncertainty; the *level* of the uncertainty—is it directly about the object or is it indirect: about the quality of the underlying evidence; or the *magnitude*?)
— What are you varying about the way you communicate the uncertainty? (The form of the *expression*, the *format* or the *medium* by which you communicate?)
— Are there aspects of the *people* doing the assessing/communicating of the uncertainty or the audience (such as their demographics, or relationship with the communicators or with the subject) that might also affect the endpoints you are interested in?

Van Der Bles, A. M., Van Der Linden, S., Freeman, A. L., Mitchell, J., Galvao, A. B., Zaval, L., & Spiegelhalter, D. J. (2019). Communicating uncertainty about facts, numbers and science. *Royal Society open science*, *6*(5), 181870.

SECTION 5

What does this mean for aiD?

SECTION 5 // What does this mean for aiD?

What does this mean for aiD

- Take trust and its mechanisms into the design of aiD systems, consider how design choices may affect trust
- Consider psychological effects of uncertainties (e.g., will they trigger otimistic bias?)
- Transparency about decision processes, limits, uncertainties
- Create a strategy to communicate uncertainties and their impact on the recommendations from an AI system
- Think about how aiD systems interact with humans and how uncertainty might affect their decision-making process



Table 11. Differences and Similarities of Metacognitions in Patterns

Pattern	Description	Distinctive metacognitive activities			Experienced conflicts	Decision accuracy
		Self-monitoring	System-monitoring	Control		
		Confirming AI Advice				
1a) Data-based confirmation	AI advice confirms patterns recognized in data	-	-	Cue accumulation	-	Typically accurate
1b) AI-based confirmation	Frame is built upon AI advice	-	-	-	-	Depends on CAID accuracy
		Disconfirming AI advice				
2a) Ignoring the AI	Ignore AI advice without further elaboration	Strong feeling of rightness	Weak belief in system capabilities	-	Belief conflict	Depends on correctness of frame
2b) Favoring the AI	Change own frame because of beliefs in CAID accuracy	Weak feeling of rightness	Strong belief in system capabilities	-	Belief conflict	Depends on CAID accuracy
3) Self-justification	Analyze data only to support own assessment	Focus on one's own decision making process	-	Frame elaboration	Validation conflict	Depends on correctness of frame
4a) Supporting the AI	Analyze data only to support AI advice	-	Develop understanding of AI	Frame elaboration	Validation conflict	Depends on CAID accuracy
4b) Explaining away	Analyze data only to find reason to reject AI advice	-	Develop understanding of AI as error	Frame elaboration	Validation conflict	Depends on correctness of frame
5) Active consideration	Neutral evaluation of all data	Focus on one's own decision making process	Develop understanding of AI	Frame elaboration+ Inference control	Validation conflict	Typically accurate

Jussupow, E., Spohrer, K., Heinzl, A., & Gawlitza, J. (2021). Augmenting medical diagnosis decisions? An investigation into physicians' decision-making process with artificial intelligence. *Information Systems Research*, *32*(3), 713-735.

PRESENTATION TITLE // Uncertainty and trust in AI systems

SPEAKER NAME // Klöckner, Christian A., Prof. Dr.

https://aid-centre.no/ DATE // 2026.09.24

